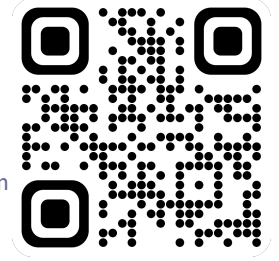


Zeel B Patel

Date of Birth: 04 Aug 1996 (29 years old)
Nationality: Indian
Website: <https://patel-zeel.github.io/>

Email (work): t-zeelpatel@microsoft.com
Email (personal): zeelpatel4896@gmail.com
GitHub: <https://github.com/patel-zeel>



EDUCATION

- **PhD in Computer Science,** *Jan 2020 - Feb 2026*
Thesis: AI for Air Quality: Modeling, Sensor Placement and Source Detection
Defended on 4th Feb 2026
Advisor: [Nipun Batra](#)
CPI: 9.7/10
IIT Gandhinagar, Gujarat, India
- **M.Tech. in Smart Manufacturing,** *Aug 2017 - May 2019*
CGPA: 9.17/10
IIITDM Kancheepuram, Chennai, India

WORK EXPERIENCE

- **PostDoc Researcher, Microsoft Research India** *Jul 2025 - Present*
- **Software Engineering Intern, Google Summer of Code** *Jun 2022 - Sep 2022*
Mentor: [Kevin P Murphy](#)
Project: Contribute code and content to code-backed Probabilistic ML textbook
GitHub repo: <https://github.com/probml/pyprobml>
Final report: <https://patel-zeel.github.io/gsoc22>
- **Data Scientist, R&D Team** *Jun 2019 - Dec 2019*
Inspirisys Solutions Ltd., Chennai, India

OPEN SOURCE LIBRARIES

BIJAX (7★): Bayesian Inference in JAX; used in Kevin Murphy's Probabilistic Machine Learning book

GPAX (6★): Gaussian processes in JAX, developed for my PhD research

skgpytorch (3★): Scikit-learn interface for GPyTorch enabling rapid GP model prototyping

ASTRA (3★): AI for Sustainability Toolkit, developed for my research group at IIT Gandhinagar

SELECTED OPEN SOURCE CONTRIBUTIONS

GPyTorch: Added probabilistic metrics module; essential for proper evaluation of probabilistic models beyond RMSE ([PR #1870](#))

supervision (Roboflow): Added Oriented Bounding Boxes support to this popular object detection library ([PR #1502](#))

matplotlib: Fixed slow plotting for PyTorch/JAX arrays; required cross-library coordination (NumPy, Matplotlib, PyTorch, JAX) ([PR #25887](#))

Stheno: Implemented FITC, one of the most widely-used sparse Gaussian process methods ([PR #17](#))

PUBLICATIONS (GOOGLE SCHOLAR PROFILE)

Journal articles

1. **Zeel B Patel**, Rishabh Mondal, Shataxi Dubey, Suraj Jaiswal, Sarath Guttikunda, Nipun Batra
Space to Policy: Scalable Brick Kiln Detection and Automatic Compliance Monitoring with Geospatial Data
Accepted in ACM Journal on Computing and Sustainable Societies and to be presented in ACM COMPASS 2025 conference.
Project page: <https://sustainability-lab.github.io/brick-kilns/>

Selected peer-reviewed articles

1. **Zeel B Patel**, Vinayak Rana, Nipun Batra
Scalable Air-Quality Sensor Placement via Gradient-Based Mutual Information Maximization
AAAI 2026 (CORE A*, Oral Presentation - 24.1% acceptance rate)
GitHub repo: <https://github.com/sustainability-lab/gdmi-aqs>
2. **Zeel B Patel**, Palak Purohit, Harsh Patel, Shivam Sahni, Nipun Batra
Accurate and Scalable Gaussian Processes for Fine-grained Air Quality Inference
AAAI 2022 (CORE A* - 15% acceptance rate)
GitHub repo: <https://github.com/patel-zeel/AAAI22>
3. Sachin Chauhan, **Zeel B Patel**, Sayan Ranu, Rijurekha Sen, Nipun Batra
Fine-Grained Spatio-Temporal Particulate Matter Dataset From Delhi For ML based Modeling
In NeurIPS 2023 Datasets and Benchmarks (CORE A* - 32.7% acceptance rate)
4. Rishiraj Adhikary, **Zeel B Patel**, Tanmay Srivasatava, Nipun Batra, Mayank Singh, Udit Bhatia, Sarath Guttikunda
Vartalaap: What Drives #AirQuality Discussions: Politics, Pollution or Pseudo-science?
CSCW 2021 (CORE A)
GitHub repo: <https://github.com/rishi-a/Vartalaap>
5. Karm Patel, Rishiraj Adhikary, **Zeel B Patel**, Nipun Batra, Sarath Guttikunda
Samachar: News Media on Air Pollution in India
COMPASS 2022
GitHub repo: <https://github.com/karm-patel/Samachar-News-media-on-air-pollution>

Symposium, Workshop papers and Posters

1. Yash Bachwana, Khush Shah, Nitish Sharma, **Zeel B Patel**, Nipun Batra, Sarath Guttikunda
VayuBuddy: LLM-powered natural language interface for exploring and understanding air pollution data
Live Demo
ACM COMPASS Posters 2024
2. Aditi Agarwal, Suraj Jaiswal, Madhav Kanda, Dhruv Patel, Rishabh Mondal, Vannsh Jani, **Zeel B Patel**, Nipun Batra, Sarath Guttikunda
Towards Scalable Identification of Brick Kilns from Satellite Imagery with Active Learning
NeurIPS Workshop on Adaptive Experimental Design and Active Learning in the Real World 2023 (CORE A*)
3. **Zeel B Patel**, Nipun Batra, Kevin Murphy
Uncertainty Disentanglement with Non-stationary Heteroscedastic Gaussian Processes for Active Learning
NeurIPS Workshop on Gaussian Processes, Spatiotemporal Modeling, and Decision-making Systems 2022 (CORE A*)
4. Aadesh Desai, Eshan Gujarathi, Saagar Parikh, Sachin Yadav, **Zeel B Patel**, Nipun Batra
Deep Gaussian Processes for Air Quality Inference
Young Researchers' Symposium, CODS-COMAD 2023
5. **Zeel B Patel***, S Deepak Narayanan*, Apoorv Agnihotri, Nipun Batra
Poster: A toolkit for spatial interpolation and sensor placement
ACM SenSys 2020 (CORE A*)
GitHub repo: <https://github.com/sustainability-lab/polire>
6. **Zeel B Patel**, Nipun Batra
Active Learning: A Visual Tour
3rd Workshop on Visualization for AI Explainability, IEEE VIS 2020 (CORE A)
Weblink: <https://patel-zeel.github.io/active-learning-visualization/>

AWARDS

Awards

- [Microsoft Research India PhD Award](#). Unrestricted grant of 1 million INR. 2024
- Outstanding Graduate Teaching Fellow award in Probabilistic Machine Learning course. Fall 2022
IIT Gandhinagar

Registration grants

- NeurIPS 2022
- [GPSS 2022](#)
- AAAI 2022
- ICML 2021
- IEEE VIS 2020

Helped advisor with

- Google PaliGemma Compute Grant 2024 (\$5000 credits in Google Cloud Platform)
- Google Compute Grant 2021 (\$5000 credits in Google Cloud Platform)

TEACHING EXPERIENCE

Graduate Teaching Fellow (teaching a course along with the instructor)

- **Probabilistic Machine Learning**
IIT Gandhinagar Fall 2022

Teaching Assistant

- **Probabilistic Machine Learning**
IIT Gandhinagar Fall 2023
- **Machine Learning**
IIT Gandhinagar Spring 2023
- **Machine Learning**
IIT Gandhinagar Spring 2022

Guest lectures

- **Introduction to Active Learning**
Ubiquitous computing, IIT Gandhinagar Fall 2021
- **Introduction to Bayesian Machine Learning**
Machine Learning, IIT Gandhinagar Spring 2021

TOOLS & SKILLS

PyTorch, JAX, Triton, Unsloth, HuggingFace Transformers, Streamlit, GeoPandas, Xarray, Docker, Git/GitHub, LaTeX, VS Code Agents (for writing ML pipelines at scale)

BOOK CONTRIBUTIONS

- **Probabilistic Machine Learning: Advanced Topics:** <https://probml.github.io/pml-book/book2.html>
I co-authored Section 34.7 (Active learning) with Dr. Kevin Murphy
- **Code-First-ML:** <https://code-first-ml.github.io/>
This book is a work-in-progress joint effort with my advisor and Prof. [Ashish Tendulkar](#) to pragmatically explain ML concepts with interactive codes and visualizations.

INVITED TALKS

Google Global Air Quality Summit

13th Nov, 2024

Topic: AirChat: LLM-Powered Chatbot System to Democratize Urban Air Quality Information

App Demo: <https://sustainabilitylabiitgn-vayubuddy.hf.space>

Gurugram, India

Air Sensors International Conference

26th Aug, 2022

Topic: Accurate and Scalable Gaussian Processes for Fine-grained Air Quality Inference

Organized by CSTEP, India and UC DAVIS

Bengaluru, India

SERVICE

Reviewer

- [Annual Conference on Neural Information Processing Systems \(CORE A*\)](#) 2024
- [Association for the Advancement of Artificial Intelligence \(AAAI\) \(CORE A*\)](#) 2024
- [Artificial Intelligence and Statistics \(AISTATS\) \(CORE A\)](#) 2023
- [ACM COMPASS Posters and Demos](#) 2021
- [The ReScience C journal](#)